APPLICATIONS OF INTEGRATION



6.1

Areas Between Curves

1 curve chapter 5

more than 1 chapter 6



In this section we learn about:

Using integrals to find areas of regions that lie between the graphs of two functions.

Definition 2

AREAS BETWEEN CURVES

The area A of the region bounded by the curves $y = f(x)$, $y = g(x)$, and the lines $x = a$, $x = b$, where f and g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$, is:

$$A = \int_a^b [f(x) - g(x)]\, dx$$

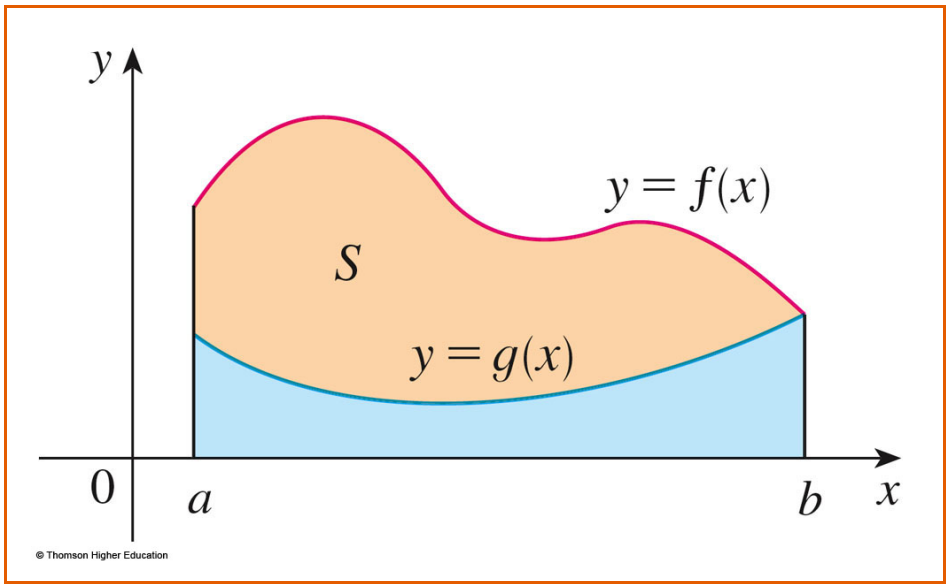
AREAS BETWEEN CURVES

- **Where both *f* and *g* are positive, you can see from the figure why Definition 2 is true:**

$$A = \left[\text{area under } y = f(x)\right] - \left[\text{area under } y = g(x)\right]$$

$$= \int_a^b f(x)dx - \int_a^b g(x)\,dx$$

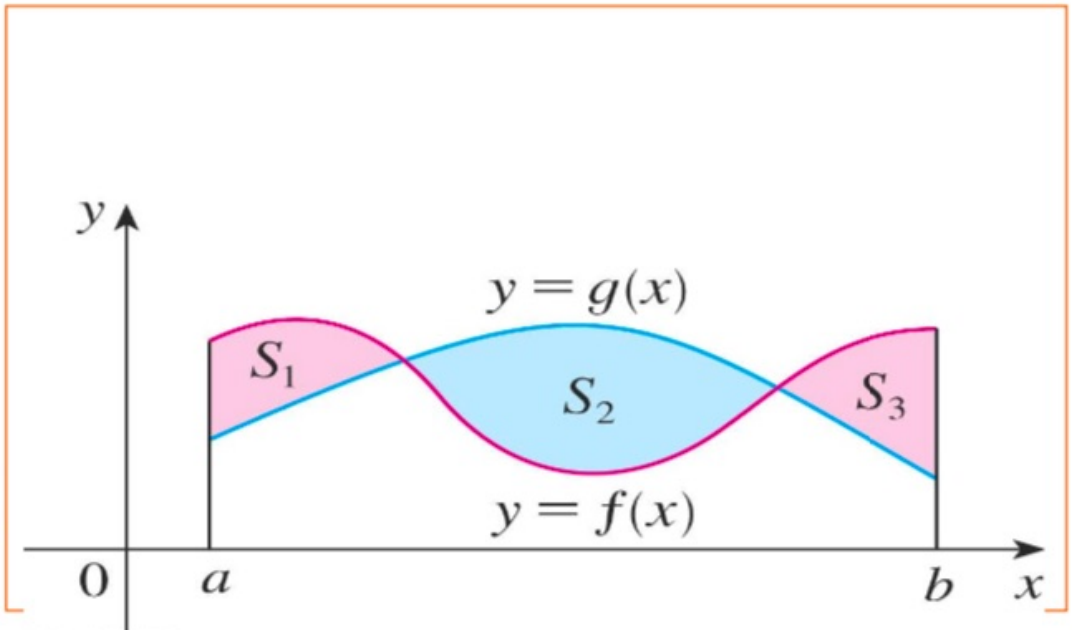
$$= \int_a^b \left[f(x) - g(x)\right]dx$$



- To find the area between the curves $y = f(x)$ and $y = g(x)$, where $f(x) \geq g(x)$ for some values of x but $g(x) \geq f(x)$ for other values of x, split the given region S into several regions S_1, S_2, . . . with areas A_1, A_2, . . . define the area of the region S to be the sum of the areas of the smaller regions S_1, S_2, . . . , that is, $A = A_1 + A_2 + \ldots$

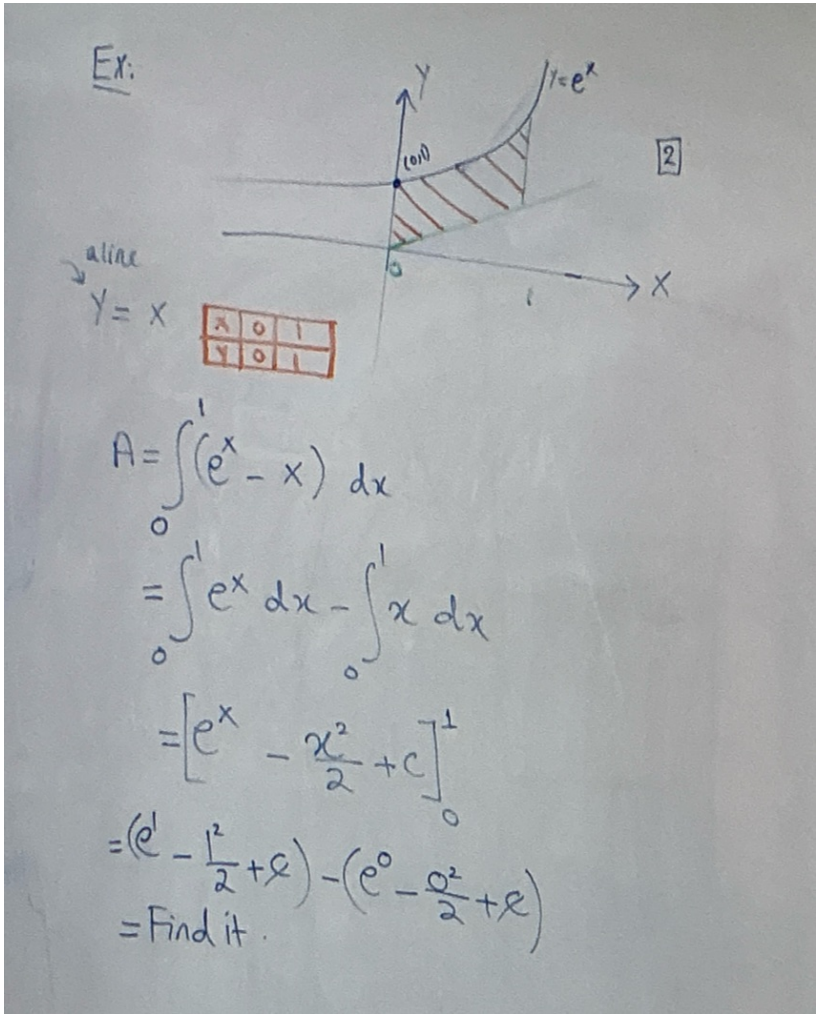
And we will use

$$\begin{cases} f(x) - g(x) & \text{when } f(x) \geq g(x) \\ g(x) - f(x) & \text{when } g(x) \geq f(x) \end{cases}$$

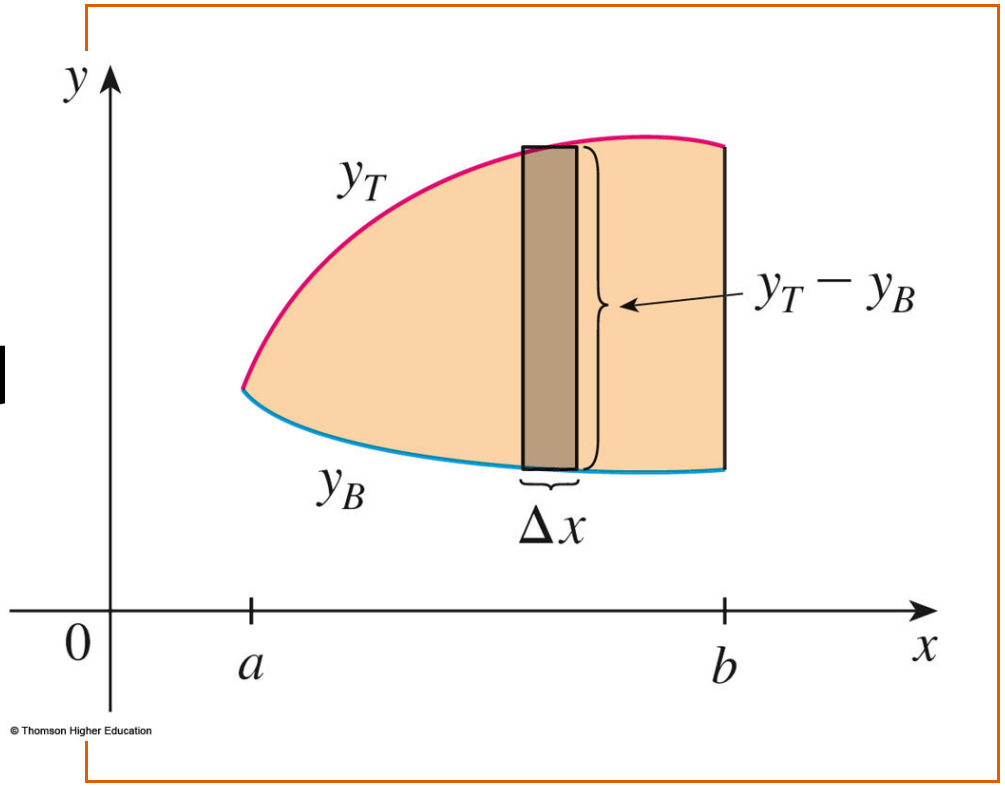


Example:

Find the area of the region bounded by $y = e^x$, and $y = x$, and bounded on the sides by $x = 0$ and $x = 1$.



- **Remark:**
- **It's helpful to sketch the region to identify the top curve *yT* , the bottom curve *yB*, and a typical approximating rectangle.**



How to find the area between two curves:

1. graph two curves.
2. Find the points of intersection of two curves which be the limit of the integration.
3. Identify which function is higher and which function is lower and then apply the rule.

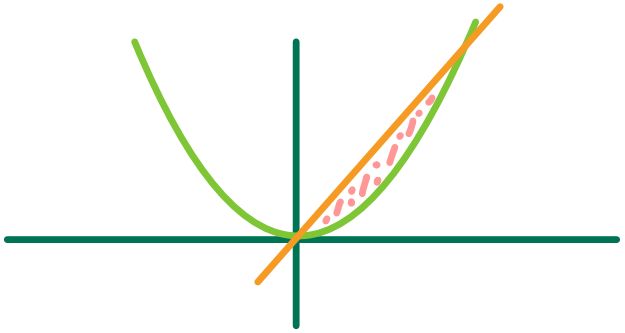
Example 3

- Find the area of the region bounded by the curves $y = \sin x$, $y = \cos x$, $x = 0$, and $x = \pi/2$.

Example 2

Find the area of the region enclosed by the parabolas $y = x^2$ and $y = x$.

y	0	1
x	0	1



$\int_a^b (\text{upper} - \text{lower})\, dx$

To find a & b, make first curve = second curve

$x^2 = x$

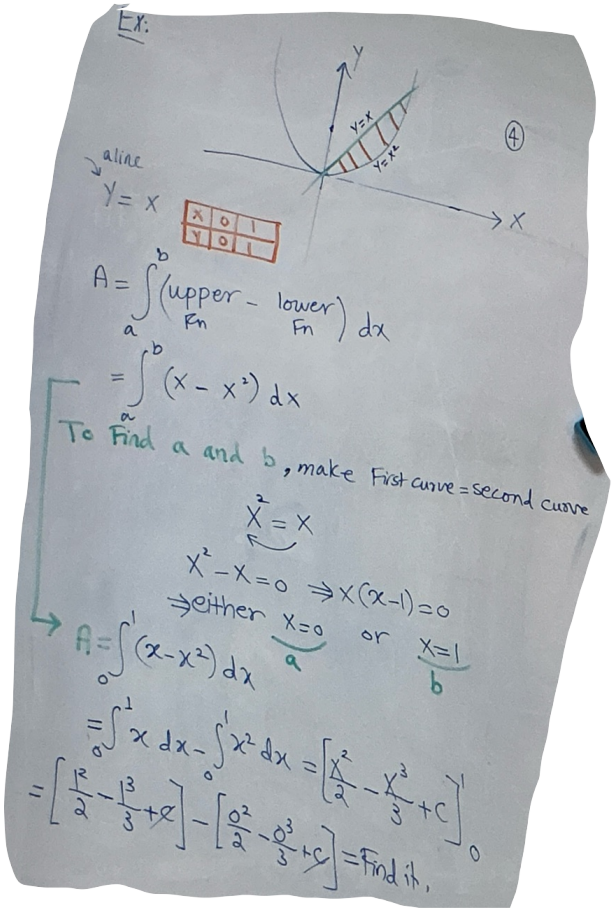
$x^2 - x = 0 \Rightarrow x(x-1) = 0$

either $x = 0$ or $x = 1$

$A = \int_0^1 (x - x^2)\, dx$

$= \int_0^1 x\, dx - \int x^2\, dx = \left[\frac{x^2}{2} - \frac{x^3}{3} + C\right]_0^1$

$= \left[\frac{1^2}{2} - \frac{1^3}{3} + C\right] - \left[\frac{0^2}{2} - \frac{0^3}{3} + C\right]$ = find it.



How to find the area between two curves:

1. graph two curves.
2. Find the points of intersection of two curves which be the limit of the integration.
3. Identify which function is higher and which function is lower and then apply the rule.

Example: Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

$x = y + 1$

Can't have a power

$2x = y^2 - 6$

$x = \frac{y^2 - 6}{2}$

Example

Find the area of the region enclosed by the parabolas

$x = y^2 - 1$ and $x = 1 - y^2$

$y = x$.

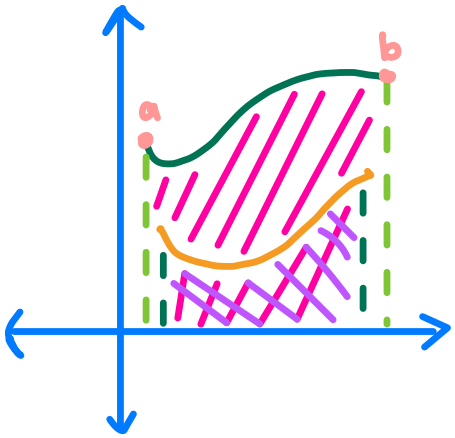
$\frac{-b}{2a}$, $f\left(\frac{-b}{2a}\right)$ → $a = 1$ $b = 0$

$y^2 - 1 = 1 - y^2$

$\frac{-0}{2(1)}$, $f\left(\frac{-0}{2(1)}\right)$ → $(0, -1)$ (y, x)

$0 = 2y^2 - 2$

$0 = 2(y^2 - 1)$



$$\text{area} = \int_{x=a}^{x=b} f(x)\,dx - \int_{x=a}^{x=b} g(x)\,dx$$

$$\int [f(x) - g(x)]\,dx = \int [\underset{\text{function}}{\text{upper}} - \underset{\text{function}}{\text{lower}}]\,dx$$

Rema rk:

Study the graphs of the following

1) $f(x) = x^2$
2) $y = x^3$
3) $y = \frac{1}{x}$, $y = \frac{1}{x^2}$
4) $y = \sqrt{x}$
5) $y = |x|$
6) $y = \cos x$, $y = \sin x$, $y = \tan x$
7) $y = \tan^{-1} x$
8) $y = e^x$
9) $y = \ln x$

Area between curves

$y = x^2 + 3x + 2$

$y = x - 1 + \cos x$

$A = \int [\text{up} - \text{low}]\, dx$

$x = \ln y + 3$

$x = e^y + y - 2$

$A = \int [\text{Right} - \text{left}]\, dy$

If we have to find the limits Make the first curve =second curve

a & b

or

c & d

$f(x) = ax^2 + bx + c$

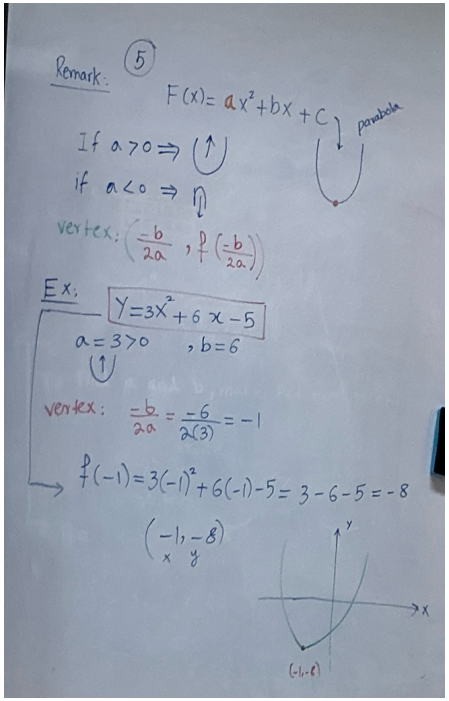
if $a > 0 \Rightarrow \cup$ Vertex : $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

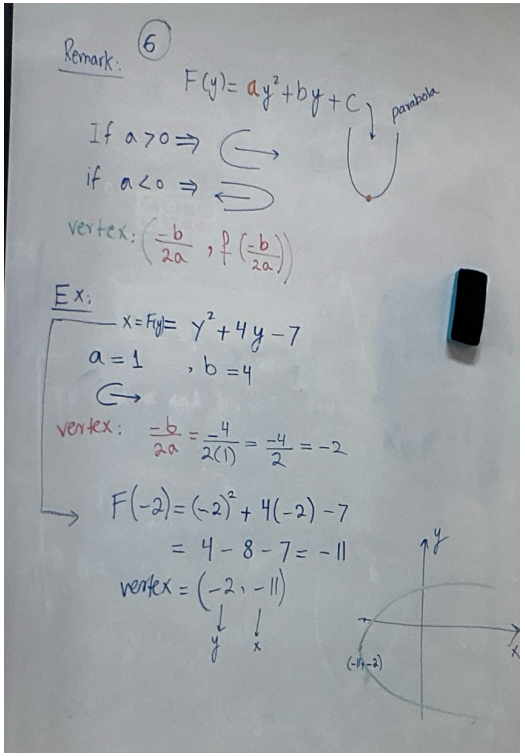
if $a < 0 \Rightarrow \cap$

$f(y) = ay^2 + by + c$

if $a > 0 \Rightarrow \subset$ Vertex : $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

if $a < 0 \Rightarrow \supset$

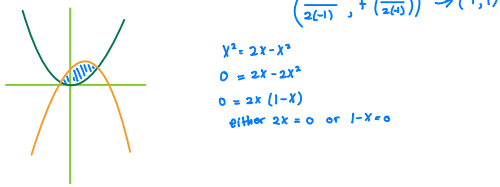




Find the area of the region enclosed by

1) $y = x^2$, $y = 2x - x^2$ $\longrightarrow$ $a = -1$ $b = 2$

$\left(\frac{-2}{2(-1)}, f\left(\frac{-2}{2(-1)}\right)\right) \rightarrow (1,1)$



$y^2 = 2x - x^2$

$0 = 2x - 2x^2$

$0 = 2x(1-x)$

either $2x = 0$ or $1 - x = 0$

$$A = \int_0^1 (2x - x^2) - x^2 \, dx = \int_0^1 2x - 2x^2 \, dx$$

$$= \int_0^1 2x \, dx - \int_0^1 2x^2 \, dx = \left(\frac{2x^2}{2} - \frac{2x^3}{3} + C\right)_0^1$$

To find a & b make first curve= second